Nathan William Henry

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Education

University of California, Berkeley 2025–present
Ph.D. in Computer Science
Advised by Stuart Russell.

University of Toronto 2020–2025
B.A.Sc. in Engineering Science (specializing in Machine Intelligence), Minor in Mathematics

- Graduated with the Engineering Science Award of Excellence.

Publications

[1] **Transformers Learn Two Competing Sorting Algorithms, and Only One Length-Generalizes**.
Khashayar Gatmiry*, **Nathan W. Henry***, Chloe H. Su, Emmanuel Abbe, Samy Bengio, Melanie Weber, Jonathan A. Kelner, Peter L. Bartlett.
Preprint, forthcoming.
*Equal contribution.

[2] **Robust Delta Networks: Improving Delta Networks with Robust Regression**.
Mayank Mishra, Bharat Runwal, Rishabh Tiwari, Samyak Jain, **Nathan W. Henry**, Tri Dao*, Joseph E. Gonzalez*.
Under review.
*Equal advising.

[3] **Geometry of Lightning Self-Attention: Identifiability and Dimension**.
Nathan W. Henry, Giovanni Luca Marchetti, Kathlén Kohn.
International Conference on Learning Representations (ICLR), 2025.
arxiv.org/abs/2408.17221

Invited Talks

- *On the Identifiability of Transformer Self-Attention.*
27th Conference of the International Linear Algebra Society (ILAS), 2026.

Service

- Program Committee member and co-organizer of the Interpretability and Science of Generalization sessions, CHAI Workshop 2026.
- Volunteer, Agentic AI Summit 2025, UC Berkeley (RDI): introduced speakers and managed session timing.